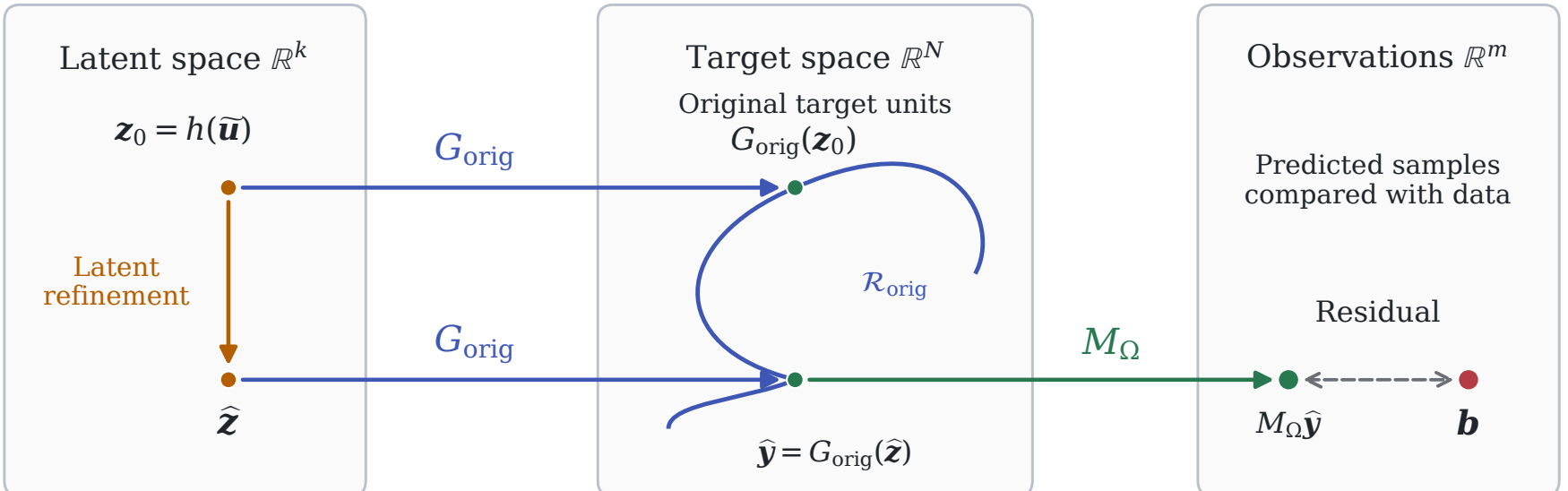

$$G_{\text{orig}}(\boldsymbol{z}) = \boldsymbol{\mu}_y + D_y G(\boldsymbol{z})$$

Refinement balances $\|M_\Omega G_{\text{orig}}(\boldsymbol{z}) - \boldsymbol{b}\|_2^2$ and $\lambda \|\boldsymbol{z} - \boldsymbol{z}_0\|_2^2$.